



EMERGENCY SOS AND INCIDENT DATA ANALYTICS SYSTEM

Real Time Assistance. Smarter Decisions



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COURSE NAME:
DATA ANALYTICS

PROJECT OVERVIEW & OBJECTIVES

OVERVIEW

An integrated system designed to manage emergency incident data and SOS reports using SQL Server and Python, with an interactive Streamlit dashboard for data analysis and visualization.

OBJECTIVES



Data Management

Collects and store emergency data and SOS reports



Pattern Identification

Identify trends across incidents, priorities, and locations.



Data Analysis

Analyze emergency data using SQL and Python.



Data Visualization

Present insights through an interactive analytics dashboard.

DATA OVERVIEW & TECHNOLOGIES

DATA OVERVIEW



Dataset: Emergency Incident Data



Records: 1+ million incident records



Stored in: SQL Server



Key Data: Call Date & Time, Priority, District Neighborhood & Incident Description



SOS Data: Emergency reports submitted through the SOS interface

TECHNOLOGIES USED



SQL Server — Data storage and SQL queries



Python — Data processing and analysis



Pandas — Data manipulation and analysis



PyODBC — Python-SQL Server connection

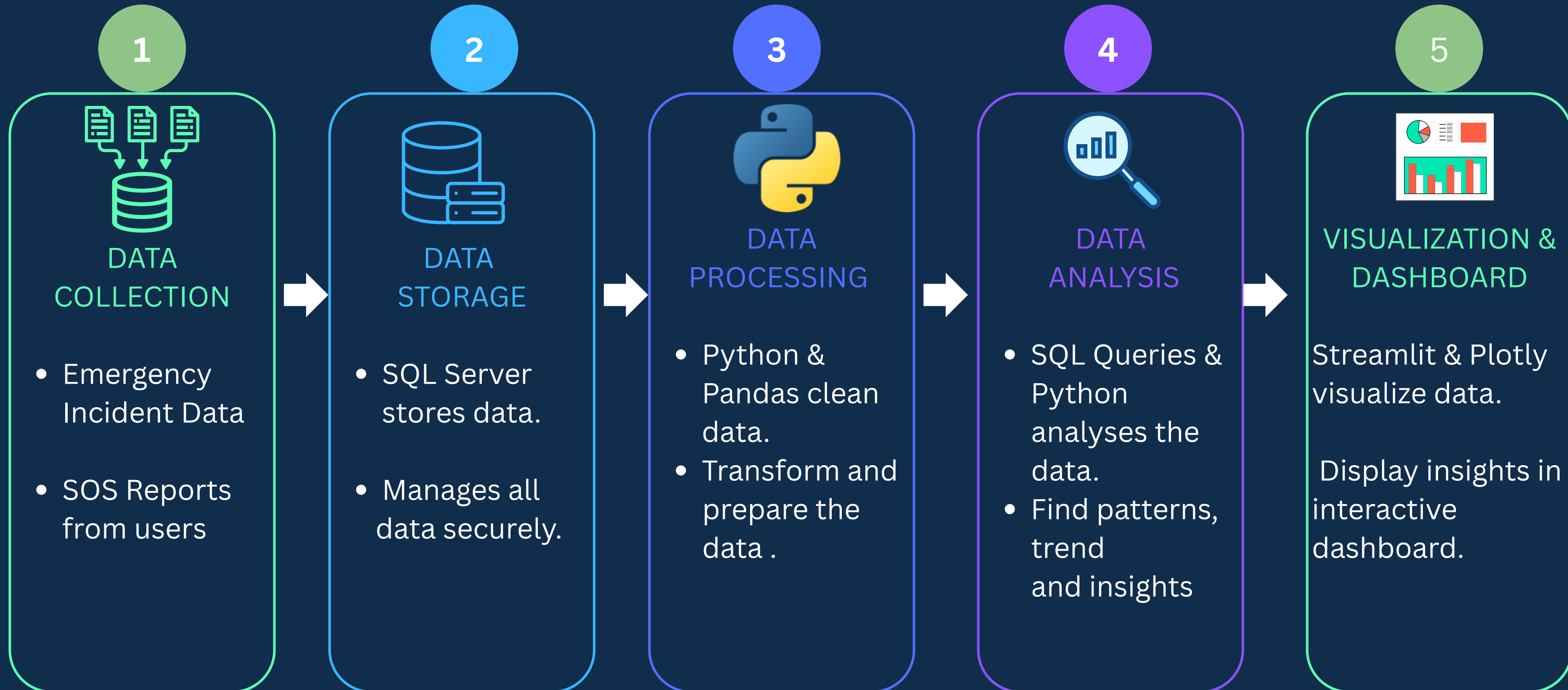


Streamlit — Web application and dashboard

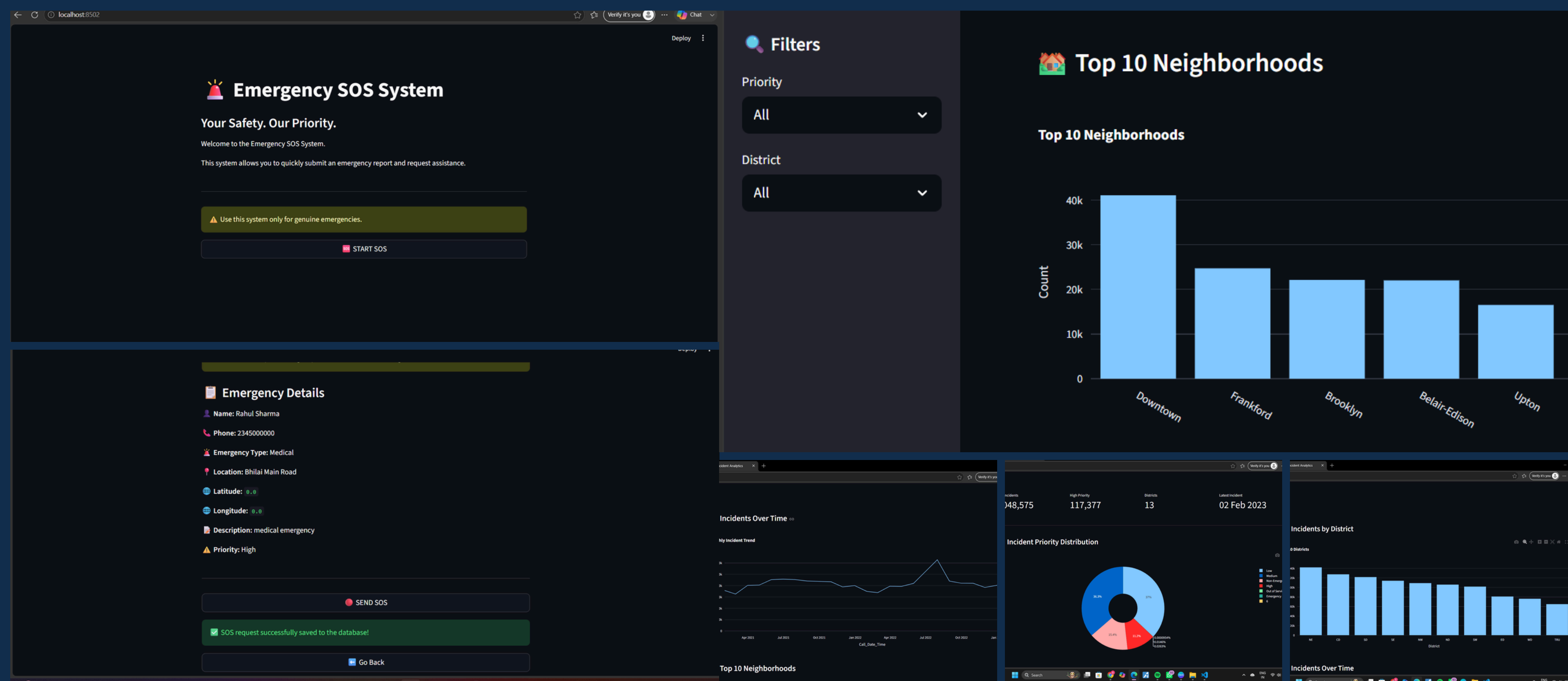


Plotly — Interactive charts and visualizations

METHODOLOGY AND SYTEM WORKFLOW



APPLICATION INTERFACE & ANALYTICS DASHBOARD



RESULTS AND KEY FINDINGS

Results

- Analyzed 1+ million incident records
- Integrated Python with SQL Server
- Developed an interactive Streamlit dashboard
- Stored SOS reports in the database

Key Findings

- Identified patterns by priority and location
- Found incident trends over time
- Visualized insights for easier decision-making

Conclusion

- The project successfully integrates SQL and Python.
- Analyzes emergency data and present meaningful insights through an interactive dashboard.

THANK YOU